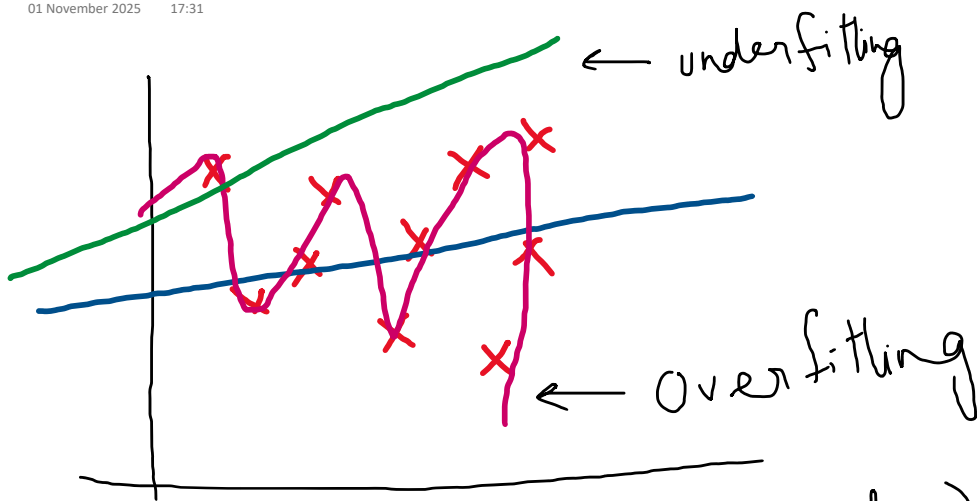




$$\hat{y} = wX + b$$

$w \leftarrow \text{vector}$
 $b \leftarrow \text{scalar}$

$w \leftarrow \text{scalar}$
 $b \leftarrow \text{scalar}$

SLR

$$w \uparrow \quad y \rightarrow \otimes \leftarrow \text{overfitting}$$

$w \leftarrow \text{array}$
 $b \leftarrow \text{scalar}$

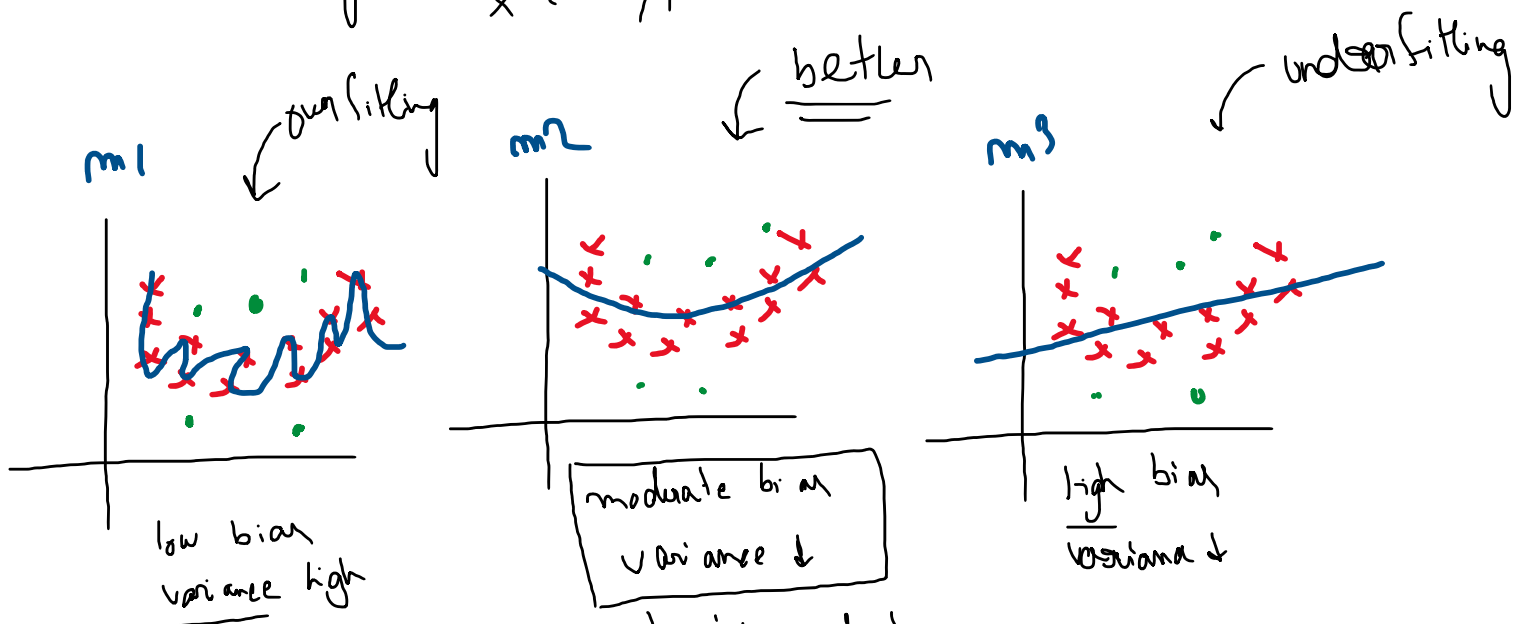
MLR

$$w \downarrow \quad y \rightarrow b \otimes \otimes$$

model did not understand the data

underfitting

$x \leftarrow \text{i/p feature}$



Bias :- Total error in training data

in testing data

Bias :-

Variance : total error in testing data

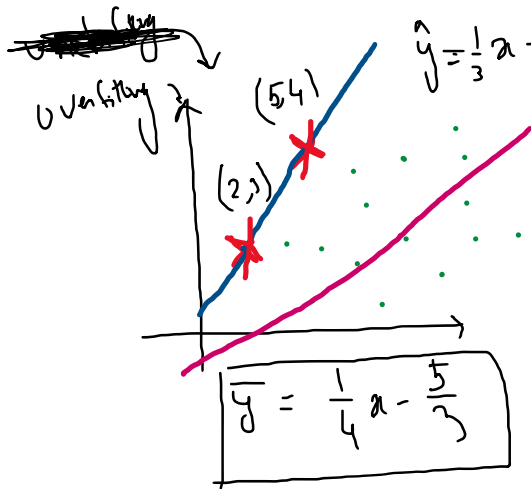
Solution:-

i) Regularization ✓

ii) Bagging

iii) Boosting

$$E_{\text{new}} = E_{\text{old}} + \text{Penalty term}$$



$$\hat{y} = \frac{1}{3}x + \frac{7}{3}$$

$$m = \frac{4-3}{5-2} = \frac{1}{3}$$

$$y = mx + b$$

$$\Rightarrow 3 = \frac{1}{3} \cdot 2 + b \quad (2,3)$$

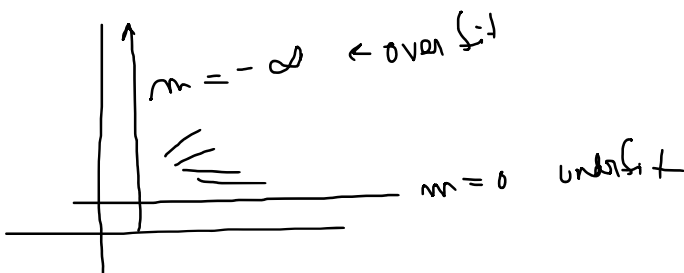
$$\Rightarrow 3 = \frac{2+3b}{3}$$

$$\Rightarrow b = \frac{7}{3}$$

$$\hat{y} = \frac{1}{3}x + \frac{7}{3}$$

$$m \downarrow$$

$$w \downarrow$$



SLR (LSS) :-

hyper parameter

$$E = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda (m^2) \quad \leftarrow \text{penalty term}$$

$$= \frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)^2 + \lambda (m^2)$$

$$E = \frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)^2 + \lambda m^2$$

$$F = \frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)^2 + \lambda m^2$$

$$\frac{dF}{db} = 0 \quad \frac{dF}{dm} = 0$$

$$\frac{dF}{db} = 0$$

$$\Rightarrow \frac{d}{db} \left[\frac{1}{n} \sum_{i=1}^n (y_i - mx_i - b)^2 + \lambda m^2 \right] = 0$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n 2(y_i - mx_i - b)(-1) + 0 = 0$$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n y_i - \frac{1}{n} \sum_{i=1}^n mx_i - \frac{1}{n} \sum_{i=1}^n b = 0$$

$$\Rightarrow \bar{y} - m\bar{x} - b = 0$$

$$\Rightarrow \boxed{b = \bar{y} - m\bar{x}} \quad \checkmark$$

$$\Rightarrow \frac{dF}{dm} = 0$$

$$\Rightarrow \frac{d}{dm} \left[\frac{1}{n} \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x})^2 + \lambda m^2 \right] = 0$$

$$\Rightarrow \frac{d}{dm} \left[\frac{1}{n} \sum_{i=1}^n (y_i - mx_i - \bar{y} + m\bar{x}) (-x_i + \bar{x}) + 2\lambda m \right] = 0$$

$$\sum_{i=1}^n$$

$$\Rightarrow \sum_{i=1}^n \left[(y_i - \bar{y}) - m(x_i - \bar{x}) \right]^2 + \lambda m^2$$

$$\Rightarrow \sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}) + m \sum_{i=1}^n (x_i - \bar{x})^2 + \lambda m^2$$

$$m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2}$$

↑
width of

$$\Rightarrow m = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2 + \lambda}$$

$$\lambda (m_1^2 + m_2^2 + m_3^2 + \dots)$$

$$L = \sum_{i=1}^n (y_i - \hat{y})^2 + \lambda (w)^2$$

$$\hat{y} = \underline{mx + b}$$

$$= (Xw - Y)^T (Xw - Y)$$

$$= (Y - Xw)^T (Y - Xw) + \lambda w^T w$$

SLR MSK

MLR:-

$$L = (Xw - Y)^T (Xw - Y) + w^T w$$

$$L = w^T X^T X w - 2w^T X^T Y + Y^T Y + w^T w$$

$$\frac{d}{dA} (A^T A) = 2(A^T)^T$$

$$L = w^T$$

$$\frac{d}{dA} (A^T \cdot A) = 2(A^T) \Rightarrow \frac{d}{dA}$$

$$\frac{dL}{dw} = 0$$

$$\frac{d}{dw} [w^T X^T X w - 2w^T X^T Y + Y^T Y + \lambda w^T w] = 0$$

$$\Rightarrow 2X^T X w - 2X^T Y + 0 + 2\lambda w = 0$$

$$\Rightarrow w(X^T X + \lambda I) = X^T Y$$

$$\Rightarrow w = (X^T X + \lambda I)^{-1} X^T Y$$

Lasso Regression

01 November 2025

18:37

L_2 / Ridge:-

Penalty term:- λw^2

L_1 / Lasso:-

Penalty term:- λw

$$\hat{y}_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + w_3 x_{i3} + \dots + w_n x_{in}$$

Lasso $\rightarrow$ feature selection $w_{\text{weights}} = 0$

bias $\uparrow$ variance $\downarrow$
sharp

Ridge $\rightarrow$ $w_{\text{weights}} \rightarrow 0$

bias $\uparrow$ variance $\downarrow$
smooth

Lasso $\rightarrow$ Higher coefficients are affected earlier.